Linux Commands

1. mount

Shows mounted filesystems, their mount points, filesystem types, and mount options.

Command:

```
mount
```

Example output / observation:

```
The screenshot shows the Linux root filesystem mounted at / and Windows drives mounted under
/mnt/c, /mnt/d, and /mnt/f.
```

Explanation: In WSL, Windows drives are commonly accessible under /mnt. The output can be long.

2. getent passwd

Retrieves user account records from configured system account databases.

Command:

```
getent passwd
```

Example output / observation:

```
Example:
root:x:0:0:root:/root:/bin/bash
Srini:x:1000:1000::/home/Srini:/bin/bash
```

Explanation: Each record follows username:password-placeholder:UID:GID:comment:home-directory:login-shell. The x indicates password data is stored separately.

3. watch -n 2 df -h

Repeats the disk-usage command every two seconds.

Command:

```
watch -n 2 df -h
```

Example output / observation:

```
The display refreshes with filesystem size, used space, available space, usage percentage, and
mount point.
```

Explanation: Press Ctrl+C to exit. If suspended with Ctrl+Z, use fg to resume it or terminate the job.

4. nc -zv localhost 8080

Tests whether a TCP connection can be made to port 8080 on the local machine.

Command:

```
nc -zv localhost 8080
```

Example output / observation:

```
Screenshot:
nc: connect to localhost (127.0.0.1) port 8080 (tcp) failed: Connection refused
```

Explanation: Connection refused usually means no service is listening on that address and port, or the connection is actively rejected.

5. blkid

Displays block-device filesystem metadata such as UUIDs and filesystem types.

Command:

```
blkid
```

Example output / observation:

```
In the screenshot, running blkid without elevated privileges did not display device entries.
```

Explanation: Some device metadata may require administrator privileges. Try sudo blkid.

6. sudo blkid

Runs blkid with administrator privileges.

Command:

```
sudo blkid
```

Example output / observation:

```
Example:
/dev/sdd: UUID="..." BLOCK_SIZE="4096" TYPE="ext4"
/dev/sdc: UUID="..." BLOCK_SIZE="4096" TYPE="swap"
```

Explanation: UUID identifies a filesystem; TYPE indicates its format. Device names and UUIDs vary. Do not copy example UUIDs into your own configuration.

7. sudo fuser -v 8080/tcp

Checks which processes are using TCP port 8080.

Command:

```
sudo fuser -v 8080/tcp
```

Example output / observation:

```
No process details were printed in the screenshot.
```

Explanation: No output commonly means no process was found using that port at the time of the check.

8. sudo lsof /var/log/syslog

Lists processes that currently have /var/log/syslog open.

Command:

```
sudo lsof /var/log/syslog
```

Example output / observation:

```
The screenshot identifies rsyslogd as a process using /var/log/syslog.
```

Explanation: lsof means list open files. Its output includes process name, PID, user, file descriptor, and file path.

9. journalctl -n 20

Displays the 20 most recent entries in the systemd journal.

Command:

```
journalctl -n 20
```

Example output / observation:

```
The screenshot includes recent messages from chronyd, snapd, systemd, and WSL-related services.
```

Explanation: Use this to inspect recent system and service events. Press q to exit the pager. WSL-specific warnings may not indicate an application failure.

10. ip addr

Displays network interfaces, their states, MAC addresses, and assigned IP addresses.

Command:

```
ip addr
```

Example output / observation:

```
Screenshot highlights:
lo: loopback interface
eth0: state UP
inet 172.31.234.174/20
```

Explanation: lo is the loopback interface. eth0 is the active interface shown. The IP address may change.

11. ip route

Displays the kernel routing table.

Command:

```
ip route
```

Example output / observation:

```
Screenshot:
default via 172.31.224.1 dev eth0 proto kernel
172.31.224.0/20 dev eth0 proto kernel scope link src 172.31.234.174
```

Explanation: The default route sends traffic for destinations without a more specific route through the gateway using eth0.

12. sudo ss -tulnp

Displays listening TCP and UDP sockets, numeric ports, and process information where available.

Command:

```
sudo ss -tulnp
```

Example output / observation:

```
The screenshot shows UDP and TCP listening sockets, including DNS-related port 53 and time-
service port 323.
```

Explanation: Options: -t TCP, -u UDP, -l listening, -n numeric addresses/ports, -p process information. Useful for checking whether a service is listening.

13. systemctl list-units --type=service

Lists service units currently loaded by systemd.

Command:

```
systemctl list-units --type=service
```

Example output / observation:

```
The screenshot lists chrony.service, console-getty.service, cron.service, dbus.service, and
other services.
```

Explanation: Columns show unit name, load state, active state, sub-state, and description. Systemd is available in the WSL environment shown.

14. lscpu

Displays CPU architecture and processor details.

Command:

```
lscpu
```

Example output / observation:

```
Screenshot highlights:
Architecture: x86_64
CPU(s): 16
Model name: 12th Gen Intel(R) Core(TM) i5-1240P
Thread(s) per core: 2
Core(s) per socket: 8
```

Explanation: Also reports CPU modes, virtualization, cache, and other processor details.

Notes:

• Port 8080 returned Connection refused, and fuser showed no process using 8080/tcp. This is consistent with no service listening on that port during the test.
• sudo blkid displayed block-device metadata, while blkid without sudo did not show entries in the screenshot.
• lsof identified rsyslogd as a process using /var/log/syslog.
• WSL exposes Windows drives through mount points such as /mnt/c, /mnt/d, and /mnt/f.